

Security Assistant – Complete Project Documentation

Live Demo: <https://security-assistant.netlify.app>

GitHub Repository: https://github.com/Karthi-1008/Security_Assistant

1. Project Overview

Security Assistant is a web-based AI-powered cybersecurity assistant designed to provide quick, structured, and practical explanations for security-related queries.

It is built as a lightweight frontend application deployed on **Netlify**, integrating an AI model API for real-time responses.

Objectives

- Provide cybersecurity explanations in simple, structured format
- Maintain clean UI and fast response
- Allow users to securely use their own API key
- Deploy as a static frontend app (no backend server required)

2. System Architecture

High-Level Architecture

User → Browser (Frontend UI) → AI API → Response → UI Display

Components

Component	Description
Frontend	HTML, CSS, JavaScript
Hosting	Netlify
AI Integration	External AI Model API
Authentication	User-provided API Key

3. Features

Core Features

- API Key Input Box (User-controlled)
- AI-based Cybersecurity Q&A
- Clean Chat Interface
- Responsive Design
- Error Handling for Invalid API
- Markdown-like formatted output

UI Features

- Modern Chat Layout
- Send Button Interaction
- Loader Animation (Optional)
- Mobile Responsive
- Clean Typography

4. Tech Stack

Layer	Technology
Frontend	HTML5
Styling	CSS3
Logic	Vanilla JavaScript
Hosting	Netlify
Version Control	Git & GitHub

5. Project Structure

Security_Assistant/

```
|
|— index.html
|— style.css
|— script.js
|— README.md
|— assets/
```

File Explanation

1 index.html

- Contains chat UI
- API key input field
- Send button
- Chat container

2 style.css

- Layout styling
- Chat bubble design
- Loader animation
- Responsive adjustments

3 script.js

- Handles:
 - API key validation
 - Fetch request to AI API
 - Error handling
 - Response formatting
 - UI rendering logic

6. Setup Instructions

♦ Run Locally

1. Clone the repository:

```
git clone https://github.com/Karthi-1008/Security_Assistant.git
```

2. Open folder:

```
cd Security_Assistant
```

3. Open index.html in browser.

♦ Deploy on Netlify

Method 1 – Drag & Drop

1. Go to Netlify Dashboard
2. Click “Add new site”
3. Drag project folder

4. Deployment completes automatically

Method 2 – GitHub Integration

1. Connect GitHub repository
2. Select repository
3. Deploy

7. How API Integration Works

Step 1: User Enters API Key

The key is stored temporarily in memory (not saved in database).

Step 2: User Sends Question

JavaScript sends POST request:

```
fetch("AI_API_ENDPOINT", {  
  method: "POST",  
  headers: {  
    "Authorization": `Bearer ${apiKey}`,  
    "Content-Type": "application/json"  
  },  
  body: JSON.stringify({  
    model: "model-name",  
    messages: [{ role: "user", content: question }]  
  })  
})
```

Step 3: Response Processing

- Parse JSON
- Extract model output
- Display inside chat UI

8. Error Handling

Error Type	Handling
Missing API Key	Alert message

Invalid API Key	Error response displayed
Network Failure	Try-catch handling
Empty Question	Validation message

9. Security Considerations

⚠ Since this is a frontend-only application:

- API key is visible in browser DevTools
- Not recommended for production without backend proxy
- For production use:
 - Implement backend server
 - Store API key securely
 - Use environment variables

10. Limitations

- No backend authentication
- API rate limit depends on user key
- No persistent chat history
- No database storage

11. Future Improvements

Suggested Enhancements

- Backend proxy (Node.js / Express)
- Secure API key handling
- Theme toggle (Dark/Light)
- Chat history saving (LocalStorage)
- Markdown rendering
- Syntax highlighting for code
- Streaming responses
- Role-based responses (Pentester Mode / Beginner Mode)

12. Use Cases

- Cybersecurity Students
- BCA / Data Science Students

- Ethical Hacking Beginners
- Quick concept revision
- Seminar preparation

13. Example Queries

- What is Man-in-the-Middle attack?
- Explain SQL Injection with example
- What is XSS?
- Difference between symmetric and asymmetric encryption
- How does ransomware work?

14. Deployment Information

Live Site:

<https://security-assistant.netlify.app>

Hosting Platform:

Netlify (Static Hosting)

15. Version Information

Version	Description
v1.0	Initial AI Chat
v1.1	UI Improvements
v1.2	Deployment Fixes

16. Author

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Conclusion

Security Assistant is a lightweight AI-based cybersecurity support tool designed for educational and learning purposes. It demonstrates:

- AI API integration
- Frontend deployment
- UI/UX implementation
- Secure API usage awareness
- Practical cybersecurity knowledge support